

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A software developer is verifying memory values before debugging a program. To validate the stored values, use the Windows Calculator (Programmer Mode) to convert the given hexadecimal values into their binary and decimal equivalents. Analyse the converted values to determine how hexadecimal simplifies binary representation and identify how an incorrect hexadecimal value could affect memory interpretation during program execution. Support your analysis with suitable conversion results.
2. A programmer wants to examine how different instructions are executed by an 8085 microprocessor. To observe the instruction execution process, write and execute an 8085 program containing data transfer and arithmetic instructions using an 8085 simulator. Observe the changes in the Program Counter (PC), Accumulator (A), and Flag Register after each instruction. Analyse how the processor performs the instruction fetch, decode, and execute stages for each instruction and explain how these stages contribute to the final output.
3. A user wants to determine which of two computers is better suited for executing the same application. To compare the system capabilities, observe the processor name, Base Speed, Current Speed, Number of Cores, Logical Processors, Installed RAM, Cache Size (if available), and System Type (32-bit/64-bit) for each system. Analyse the observed specifications to determine which computer is expected to execute the application more efficiently, and justify your conclusion by relating the processor and memory characteristics to overall system performance.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge